

Sebastian F. Otarola-Bustos, PhD

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Professional Summary

I am an atmospheric and hydrologic scientist. I hold a PhD in Fluid Dynamics and have 4 years of industry experience in water modeling and data science. For my PhD, I participated in two major field campaigns on the U.S. west coast, where I operated scientific instrumentation to study atmospheric flows over complex terrain to improve boundary layer parameterizations in weather models.

At Hazen, I specialize in hydraulic and hydrologic modeling, GIS analysis, and Machine Learning (ML) for predictive modeling in flood and water supply projects. Outside of my role at Hazen, I pursue independent applied research projects and academic collaborations. For more information on my work, please visit my personal website.

Education

PhD, Civil & Environmental Engineering & Earth Sciences

University of Notre Dame – Notre Dame, IN | 2016 –2022

B.S., Water Resources Engineering

Pontificia Universidad Católica de Chile – Santiago, Chile | 2009 –2015

Technical Skills

Programming & Computational Stack

Python Programming • MATLAB • GitHub Workflows • SQL, SQLite, PostgreSQL • Docker • C++ • Linux • HTML • Cloud Computing • LLMS • Agentic AI Workflows • ArcGIS Pro • H&H Modeling Software

Atmospheric & Climate Models

NWP Forecasting Systems (GFS, ECMWF IFS/AIFS, NAM, HRRR, WRF) • CMIP6 Project • ERA 5 • Remote Sensing (Sentinel 1, Sentinel 2, GOES)

Professional Experience

Hazen – Principal Scientist

Silver Spring, MD | Aug 2022 – Present

- Developed and calibrated hydrologic models for utilities and water districts across the US. The models were used to meet regulatory compliance in topics such as flood analysis, water supply management, evaluation of capital improvement projects, and flood analysis in urban and natural systems.
- Contributed to a framework for building and calibrating parsimonious (sparse-parameter) hydrological models in Python, aimed at flooding during wet-weather events in stormwater systems.
- Lead machine learning (ML) initiatives in classification, regression, and anomaly detection including flow forecasting tool, urban flooding detection, and water demand forecasting. This work followed best software development practices in GitHub and leveraged modern ML frameworks.
- Led data analysis and tool development for Water Research Foundation (WRF) project on water quality in quarry lakes. The project leveraged the LAGOS dataset that included information retrieved from in-situ and remote sensing measurements for about 500,000 lakes in the US. A web tool was developed to screen water quality in quarry lakes and is available to WRF subscribers.

Research & Independent Projects

2025–Present

- I am training a convolutional neural network (CNN) to downscale CMIP6-based climate projections of extreme precipitation and temperature in Chile. The CNN can capture Chile's unique topography, and we are investigating the physical mechanisms driving the changes. This is part of a manuscript in preparation for *Climate Dynamics*.
- I am evaluating Long Short-Term Memory (LSTM) and diffusion algorithms to predict flow discharges in U.S. west coast Basins during Atmospheric River (AR) events. I am benchmarking the results against process-based models (GR4J and CemaNeige) for various basins' hydrological regimes. This work is part of a manuscript in preparation for the *Journal of Hydrology*.
- I maintain two weather intelligence platforms with live data ingestion from operational weather forecasting models and observational weather data and a live bias-correction pipeline to calibrate the forecasts: 1) A U.S. heat-wave tracker for all major airports in the U.S., and 2) A storm tracker that monitors atmospheric rivers, weather forecasts, weather station data, and flood alerts in Chile.

Research Experience

University of Notre Dame – PhD Researcher, Atmospheric Fluid Dynamics

Notre Dame, IN | Aug 2016 – May 2022

- Participated in two major experiments including the Wind Forecast Improvement Project 2 (WFIP2) in the Columbia River (OR) and the Sundowner Wind Experiment (SWEX) in Santa Barbara (CA). I deployed flux towers and doppler lidars and processed their observations to study the skill of weather forecasting models in complex terrain.
- Published results associated with the WFIP2 experiment in Boundary Layer Meteorology, where I studied the sub-grid variability of surface layer turbulence in complex terrain using 18-month observational data, and benchmarked the results against the WRF model.
- Collaborated in peer-reviewed publications in boundary layer and wind energy; produced open datasets published in the DOE Atmosphere to Electrons Archive and presented research in national and international conferences.

Papers

- **Otarola-Bustos, S.F.**, Giani, P., & Valenzuela, R. et al. (2027). LSTM and diffusion models compared with process-based and operational benchmarks during atmospheric-river floods in US West Coast basins. *Journal of Hydrology*. (*in preparation*).
- **Otarola Bustos, S. F.**, & Buahin, C. (2027). Physical Drivers of Downscaled CMIP6 Projections of weather extremes over Chile. *Climate Dynamics* (*in preparation*).
- **Otarola-Bustos, S. F.** et al. (2023). Subgrid variability of surface-layer parameters in complex terrain. *Boundary-Layer Meteorology*, 187, 229–265.
- Grachev, A. et al. (2022). Hybrid bulk algorithm for turbulent fluxes over dry and wet soils. *Journal of Applied Meteorology and Climatology*, 61, 393–414.
- Grachev, A. et al. (2020). On the surface energy balance closure at different temporal scales, *Agricultural and Forest Meteorology*.
- Wilczak, J. M. et al. (2019); Fernando, H. J. S. et al. (2019). WFIP2 and Perdigão field campaign overviews. *Bulletin of the American Meteorological Society*.
- Fernando, H. J. et al. (2018). *The Perdigao: Peering into microscale details of mountain winds*, *B. Am. Meteorol. Soc.*, 100, 799–820.
- Wilcox, A. C. et al. (2016). Integrated analysis of the March 2015 Atacama floods. *Geophysical Research Letters*, 43, 8035–8043.

Conferences

- **Otarola Bustos, S. F.**, Giani, P., & Valenzuela, R. (2026). Downscaling CMIP6 projections of weather extremes over Chile: A CNN super-resolution approach. Abstract submitted in Session A060: Decision-Relevant Understanding of Impactful Weather and Extremes, AGU Fall Meeting 2026, San Francisco, CA, 7–11 Dec.
- **Otarola Bustos, S. F.**, & Buahin, C. (2026). LSTM and diffusion models compared with process-based and operational benchmarks during atmospheric-river floods in US West Coast basins. Abstract submitted in Session H039: AI-Driven Hydrologic Modeling and Prediction: Methods, Applications, and Emerging Frontiers, AGU Fall Meeting 2026, San Francisco, CA, 7–11 Dec.
- **Otarola Bustos, S. F.** Measurements of surface layer parameters in complex terrain during the WFIP experiment in the Columbia River Gorge (OR). AGU Fall Meeting 2022, Chicago, IL, 12-16 Dec.
- Duine, G. J., Carvalho, L. M. V., De Wekker, S. F. J., Jones, C., de Orla-Barile, M., Whang, Z., Kasić, J., Fernando, H. J. S., & **Otarola Bustos, S. F.** (2022). Exploring the spatiotemporal variability of lee slope jets during the Sundowner Winds Experiment. 20th Conference on Mountain Meteorology, American Meteorological Society, Park City, UT, June 2022.

Teaching & Mentoring

Engineering Mentoring – Hazen and Sawyer

Aug 2022 – Present

- Mentored junior engineers in hydraulic modeling (InfoWater Pro, PCSWMM), GIS software, and Python-based workflows to automate labor-intensive tasks and increase work accuracy.
- Trained junior engineers and scientist in GitHub workflows and best development practices in collaborative data science and machine learning projects.

Graduate Teaching Assistant - University of Notre Dame

2017 – 2018

- **CE 30455: Environmental Hydrology** (Fall 2017, Fall 2018). Conducted office hours and problem-solving sessions ahead of mid-terms. Graded homework's, mid-terms, and exams. Prepared laboratory demonstrations (Prof. Andrew Kennedy).
- **CE 30460 Fluid Mechanics** (Spring 2017, Spring 2018). Conducted office hours and problem-solving sessions ahead of mid-terms. Graded homework's, mid-terms, and exams (Prof. David Richter).